



Recitation 6

Problem Set 6 (Problems 5~9)
Problem Set 7

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O U T L I N E

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01

PART ONE

Problem Set 6

1.1 Problem 5

- OWN Problem 5.19

5.19. Consider a causal and stable LTI system S whose input $x[n]$ and output $y[n]$ are related through the second-order difference equation

$$y[n] - \frac{1}{6}y[n-1] - \frac{1}{6}y[n-2] = x[n].$$

- (a) Determine the frequency response $H(e^{j\omega})$ for the system S .
- (b) Determine the impulse response $h[n]$ for the system S .



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1.1 Problem 5

- (a) Taking the Fourier transform of both sides of the difference equation, we have

$$Y(e^{j\omega}) - \frac{1}{6}e^{-j\omega}Y(e^{j\omega}) - \frac{1}{6}e^{-j2\omega}Y(e^{j\omega}) = X(e^{j\omega})$$

- Therefore

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{1}{1 - \frac{1}{6}e^{-j\omega} - \frac{1}{6}e^{-j2\omega}} = \frac{1}{\left(1 - \frac{1}{2}e^{-j\omega}\right)\left(1 + \frac{1}{3}e^{-j\omega}\right)}$$

- If you were sensitive enough to notice that $x^2 + x - 6 = (x - 2)(x + 3)$.
- (b) Using partial fraction expansion

$$H(e^{j\omega}) = \frac{1}{\left(1 - \frac{1}{2}e^{-j\omega}\right)\left(1 + \frac{1}{3}e^{-j\omega}\right)} = \frac{3/5}{\left(1 - \frac{1}{2}e^{-j\omega}\right)} + \frac{2/5}{\left(1 + \frac{1}{3}e^{-j\omega}\right)}$$

- Taking the inverse Fourier transform of both sides, we have

$$h[n] = \frac{3}{5}\left(\frac{1}{2}\right)^n u[n] + \frac{2}{5}\left(-\frac{1}{3}\right)^n u[n]$$

- Don't forget $u[n]$ as the system is causal.

- OWN Problem 5.34

5.34. Consider a system consisting of the cascade of two LTI systems with frequency responses

$$H_1(e^{j\omega}) = \frac{2 - e^{-j\omega}}{1 + \frac{1}{2}e^{-j\omega}}$$

and

$$H_2(e^{j\omega}) = \frac{1}{1 - \frac{1}{2}e^{-j\omega} + \frac{1}{4}e^{-j2\omega}}.$$

- Find the difference equation describing the overall system.
- Determine the impulse response of the overall system.



- (a) Since the two systems are cascaded, the frequency response of the overall system is

$$H(e^{j\omega}) = H_1(e^{j\omega})H_2(e^{j\omega}) = \frac{2 - e^{-j\omega}}{1 + \frac{1}{8}e^{-j3\omega}}$$

- You can use the **cube sum formula** $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$ directly on the denominator instead of expanding the calculation (of course, the direct calculation is not wrong). Therefore

$$Y(e^{j\omega}) + \frac{1}{8}e^{-j3\omega}Y(e^{j\omega}) = 2X(e^{j\omega}) - e^{-j\omega}X(e^{j\omega})$$

- Taking the inverse Fourier transform of both sides, we get the difference equation

$$y[n] + \frac{1}{8}y[n-3] = 2x[n] - x[n-1]$$



- (b) We may write overall frequency response as

$$H(e^{j\omega}) = \frac{4/3}{1 + \frac{1}{2}e^{-j\omega}} + \frac{(1 + j\sqrt{3})/3}{1 - \frac{1}{2}e^{j\pi/3}e^{-j\omega}} + \frac{(1 - j\sqrt{3})/3}{1 - \frac{1}{2}e^{-j\pi/3}e^{-j\omega}}$$

- Taking the inverse Fourier transform of both sides, we obtain

$$h[n] = \frac{4}{3} \left(-\frac{1}{2}\right)^n u[n] + \frac{1 + j\sqrt{3}}{3} \left(\frac{1}{2}e^{j\pi/3}\right)^n u[n] + \frac{1 - j\sqrt{3}}{3} \left(\frac{1}{2}e^{-j\pi/3}\right)^n u[n]$$

- You can also write it in cosine form.

$$h[n] = \frac{4}{3} \left(-\frac{1}{2}\right)^n u[n] + \frac{4}{3} \cos\left(\frac{\pi}{3}(n+1)\right) \left(\frac{1}{2}\right)^n u[n]$$

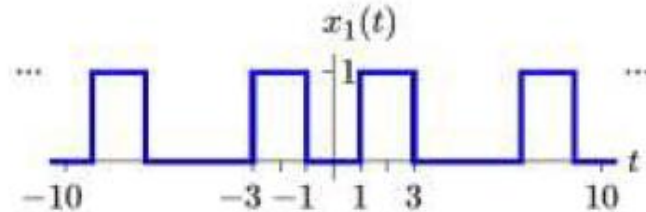


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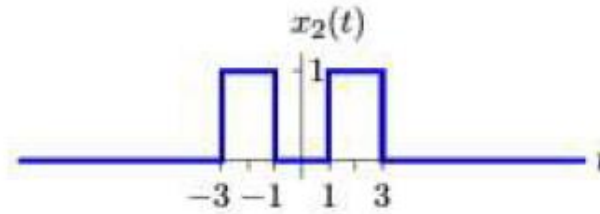
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1.3 Problem 7

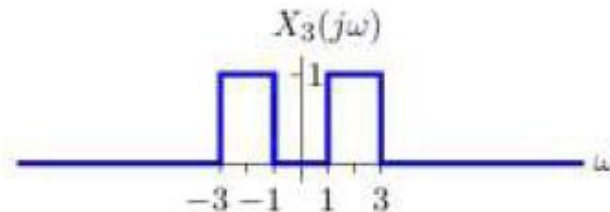
- a. Determine the Fourier series coefficients of the following signal, which is periodic in $T = 10$.



- b. Determine the Fourier transform of the following signal, which is zero outside the indicated range.



- c. Determine the time waveform that corresponds to the following Fourier transform, which is zero outside the indicated range.



What is the relation between the answer to this part and that of the previous part?



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1.3 Problem 7

- (a) For $x_1(t)$, we obtain $T = 10$, so $\omega_0 = 2\pi/10 = \pi/5$. From (3.37), the FS coefficients are

$$\begin{aligned} a_k &= \frac{1}{10} \int_{-3}^{-1} e^{-jk(\frac{\pi}{5})t} dt + \frac{1}{10} \int_1^3 e^{-jk(\frac{\pi}{5})t} dt = \frac{e^{-jk(\frac{3\pi}{5})} - e^{-jk(\frac{\pi}{5})} + e^{jk(\frac{\pi}{5})} - e^{jk(\frac{3\pi}{5})}}{-j2k\pi} \\ &= \frac{1}{\pi k} \left(\frac{e^{jk(\frac{3\pi}{5})} - e^{-jk(\frac{3\pi}{5})}}{2j} - \frac{e^{jk(\frac{\pi}{5})} - e^{-jk(\frac{\pi}{5})}}{2j} \right) = \frac{1}{\pi k} \left(\sin \frac{3k\pi}{5} - \sin \frac{k\pi}{5} \right) \end{aligned}$$

- Don't forget to calculate $a_0 = \frac{1}{10} \int_{-3}^{-1} dt + \frac{1}{10} \int_1^3 dt = \frac{2}{5}$.
- (b) You can calculate directly

$$\begin{aligned} X_2(j\omega) &= \int_{-\infty}^{\infty} x_2(t) e^{-j\omega t} dt = \int_{-3}^{-1} e^{-j\omega t} dt + \int_1^3 e^{-j\omega t} dt \\ &= \frac{e^{j\omega} - e^{j3\omega} + e^{-j3\omega} - e^{-j\omega}}{-j\omega} = \frac{2(\sin 3\omega - \sin \omega)}{\omega} \end{aligned}$$

- Don't forget to calculate $X_2(j0) = 2 \int_1^3 dt = 4$.

- (c) You can calculate directly

$$\begin{aligned}x_3(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X_3(j\omega) e^{j\omega t} d\omega = \frac{1}{2\pi} \int_{-3}^{-1} e^{j\omega t} d\omega + \frac{1}{2\pi} \int_1^3 e^{j\omega t} d\omega \\&= \frac{e^{-jt} - e^{-j3t} + e^{j3t} - e^{jt}}{2j\pi t} = \frac{\sin 3t - \sin t}{\pi t}\end{aligned}$$

- Don't forget to calculate $x_3(0) = 2 \times \frac{1}{2\pi} \int_1^3 d\omega = \frac{2}{\pi}$.
- The answer to this question is dual to the previous one. (You'll lose a little point if you don't write the relationship.)

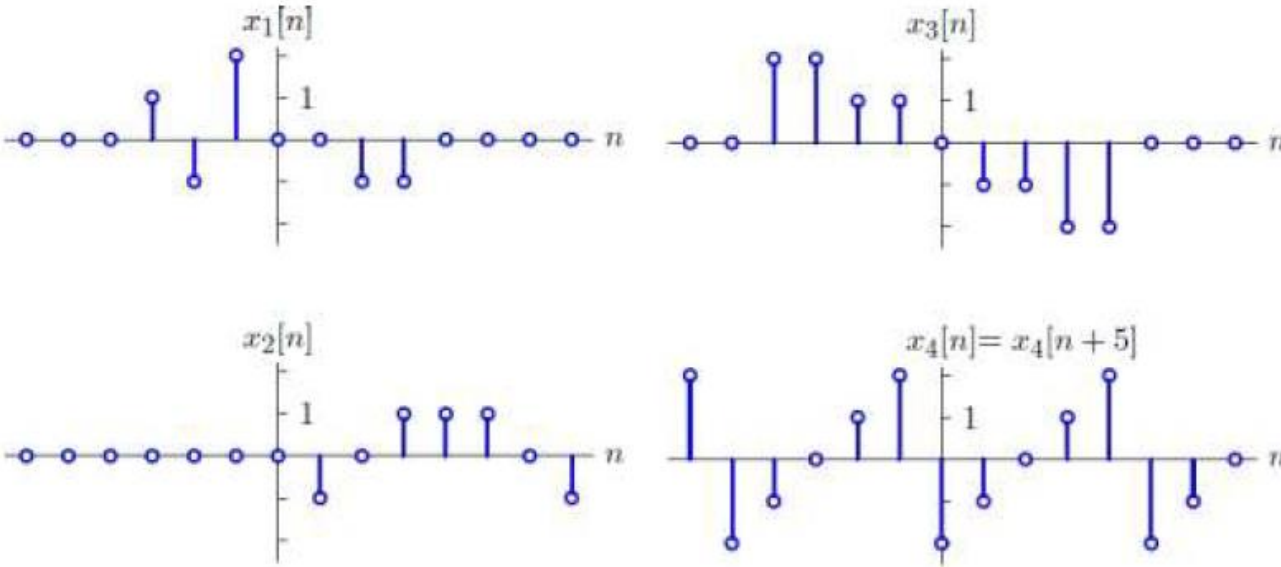




Which are True?

For each of the DT signals $x_1[n]$ through $x_4[n]$ (below), determine whether the conditions listed in the following table are satisfied, and answer **T** for true or **F** for false.

	$x_1[n]$	$x_2[n]$	$x_3[n]$	$x_4[n]$
$X(e^{j0}) = 0$				
$\int_{-\pi}^{\pi} X(e^{j\Omega}) d\Omega = 0$				
$X(e^{j\Omega})$ is purely Imaginary				
$e^{jk\Omega}X(e^{j\Omega})$ is purely real for some integer k				



1.4 Problem 8

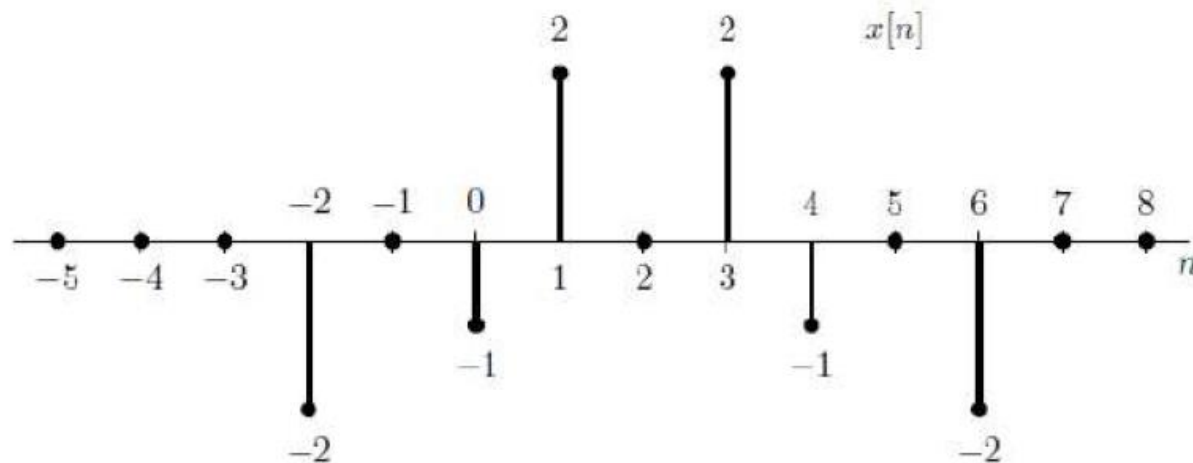
- Analyze each of the four conditions given in the question.
- (1) $X(e^{j0}) = 0$ means $\sum_{n=-\infty}^{\infty} x[n] = 0$, then $x_1[n]$, $x_3[n]$ and $x_4[n]$ meet this condition, while $\sum_{n=-\infty}^{\infty} x_2[n] = 1$. (Note: for $x_4[n]$, you may use (5.19) to (5.20) to calculate $X(e^{j0})$, or you will get divergent results. We generally think that the product of 0 and an impulse function is 0.)
- (2) $\int_{-\pi}^{\pi} X(e^{j\Omega}) d\Omega = 0$ means $x[0] = 0$, then $x_1[n]$, $x_2[n]$ and $x_3[n]$ meet this condition, while $x_4[0] = -2$.
- (3) $X(e^{j\Omega})$ is purely imaginary, then $x[n]$ should be real and odd, only $x_3[n]$ meets this condition.
- (4) $e^{jk\Omega} X(e^{j\Omega})$ is purely real for some integer k , then $x[n+k]$ should be real and even, only $x_2[n]$ meets this condition.



	$x_1[n]$	$x_2[n]$	$x_3[n]$	$x_4[n]$
$X(e^{j0}) = 0$	T	F	T	T
$\int_{-\pi}^{\pi} X(e^{j\Omega}) d\Omega = 0$	T	T	T	F
$X(e^{j\Omega})$ is purely imaginary	F	F	T	F
$e^{jk\Omega} X(e^{j\Omega})$ is purely real for some integer k	F	T	F	F



Let $X(e^{j\omega})$ denote the Fourier transform of the signal $x[n]$ depicted below.



- Find $X(1) = X(e^{j0})$.
- Find α such that $e^{j\alpha\omega} X(e^{j\omega})$ is real.
- Evaluate $\int_{-\pi}^{\pi} X(e^{j\omega}) d\omega$.
- Find $X(e^{j\pi})$.
- Determine and sketch the signal whose Fourier transform is $\Re\{X(e^{j\omega})\}$.
- Evaluate the following integrals:

$$\int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$



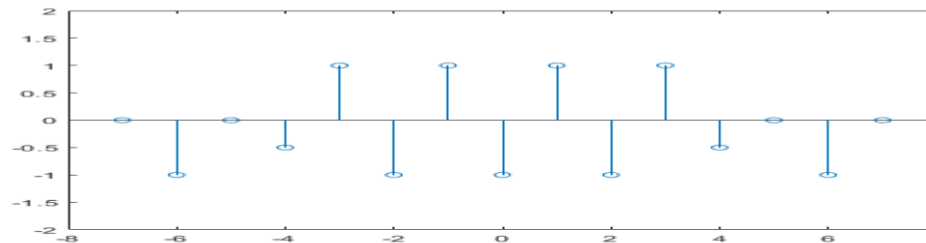
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1.5 Problem 9

- (a) $X(e^{j0}) = \sum_{n=-\infty}^{\infty} x[n] = -2 - 1 + 2 + 2 - 1 - 2 = -2$.
- (b) As $e^{j\alpha\omega}X(e^{j\omega})$ is purely real, we require $x[n + \alpha]$ should be real and even, then $\alpha = 2$ (since the original $x[n]$ is symmetric about 2).
- (c) $\int_{-\pi}^{\pi} X(e^{j\omega})d\omega = 2\pi x[0] = 2\pi \times (-1) = -2\pi$.
- (d) $X(e^{j\pi}) = \sum_{n=-\infty}^{\infty} e^{-j\pi n}x[n] = \sum_{n=-\infty}^{\infty} (-1)^n x[n] = x[-2] + x[0] - x[1] - x[3] + x[4] + x[6] = -2 - 1 - 2 - 2 - 1 - 2 = -10$.
- (e) The inverse Fourier transform of $\mathcal{R}e\{X(e^{j\omega})\}$ is $\mathcal{E}v\{x[n]\} = (x[n] + x[-n])/2$. Then the signal is like



- (f) Using Parseval's relation

$$\int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega = 2\pi \sum_{n=-\infty}^{\infty} |x[n]|^2 = 2\pi [(-2)^2 + (-1)^2 + 2^2 + 2^2 + (-1)^2 + (-2)^2] = 36\pi$$

02

PART TWO

Problem Set 7

2.1 Problem 1

1. Sampling CT sinusoids

Consider 3 CT signals:

$$x_1(t) = \cos(3000t) ,$$

$$x_2(t) = \cos(4000t) , \text{ and}$$

$$x_3(t) = \cos(5000t) .$$

Each of these is sampled as follows

$$x_1[n] = x_1(nT) ,$$

$$x_2[n] = x_2(nT) , \text{ and}$$

$$x_3[n] = x_3(nT) ,$$

where $T = 0.001$. Which of the resulting DT signals has the highest DT frequency?
Which has the lowest DT frequency?



2.1 Problem 1

- This problem is on slide Lec14.

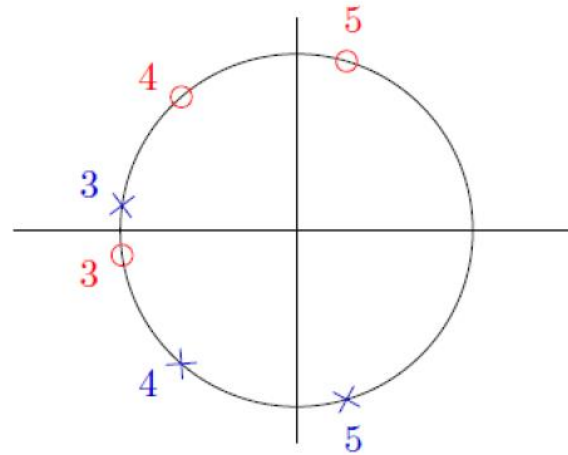
The discrete signals are

$$x_1[n] = \cos[3n]$$

$$x_2[n] = \cos[4n]$$

$$x_3[n] = \cos[5n]$$

and the corresponding discrete frequencies are $\Omega = 3, 4$ and 5 , represented below with $\times$ marking $e^{j\Omega}$ and $\circ$ marking $e^{-j\Omega}$.



- In fact, $\cos[4n] = \cos[(2\pi - 4)n]$, $\cos[5n] = \cos[(2\pi - 5)n]$. As $2\pi \approx 6.28$, $2\pi - 4 \approx 2.28 < 3$, $2\pi - 5 \approx 1.28 < 3$. Then $x_1[n]$ has the highest DT frequency, while $x_3[n]$ has the lowest DT frequency. Aliasing occurs when the last two CT signals are sampled.



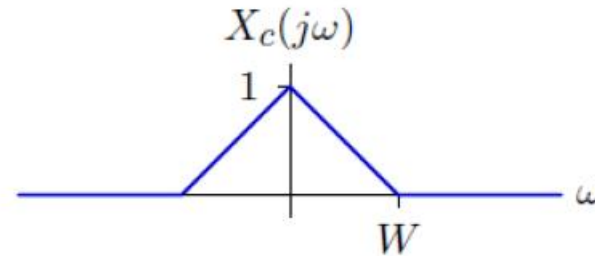
2.2 Problem 2

2. Sampling with alternating impulses

A CT signal $x_c(t)$ is converted to a DT signal $x_d[n]$ as follows:

$$x_d[n] = \begin{cases} x_c(nT) & n \text{ even} \\ -x_c(nT) & n \text{ odd} \end{cases}$$

- a. Assume that the Fourier transform of $x_c(t)$ is $X_c(j\omega)$ shown below.



Determine the DT Fourier transform $X_d(e^{j\Omega})$ of $x_d[n]$.

- b. Assume that $x_c(t)$ is bandlimited to $-W \leq \omega \leq W$. Determine the maximum value of W for which the original signal $x_c(t)$ can be reconstructed from the samples $x_d[n]$.
- c. Make a diagram of a system to reconstruct $x_c(t)$ from $x_d[n]$.



2.2 Problem 2

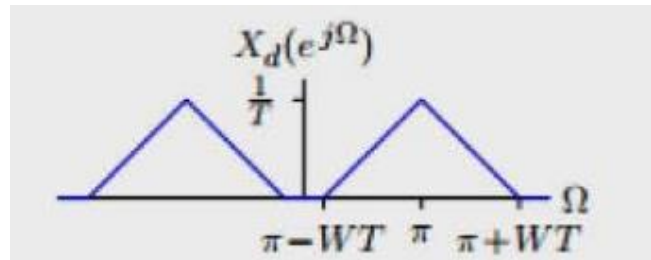
- (a) Let $x_e[n] = x_c(nT)$, then

$$X_e(e^{j\Omega}) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} X_c\left(j\left(\frac{\Omega - 2k\pi}{T}\right)\right)$$

- And

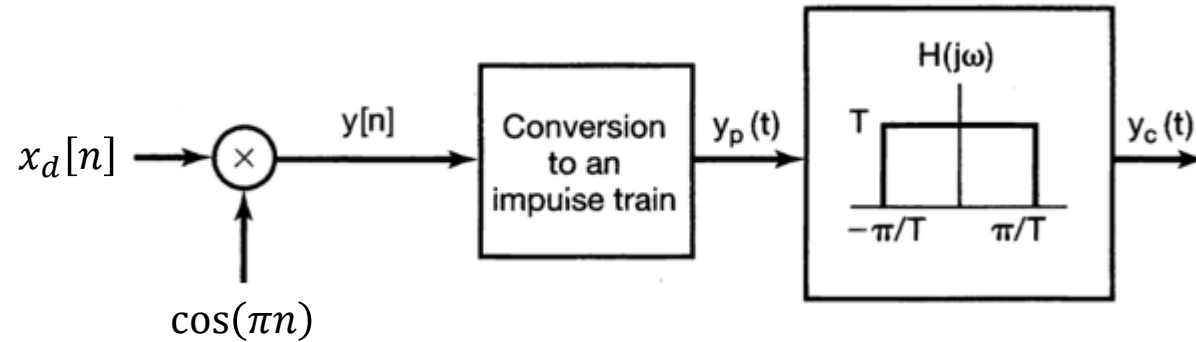
$$\begin{aligned} x_e[n] &= (-1)^n x_d[n] = (-1)^n \frac{1}{2\pi} \int_{2\pi} X_d(e^{j\Omega}) e^{j\Omega n} d\Omega = e^{j\pi n} \frac{1}{2\pi} \int_{2\pi} X_d(e^{j\Omega}) e^{j\Omega n} d\Omega \\ &= \frac{1}{2\pi} \int_{2\pi} X_d(e^{j\Omega}) e^{j(\Omega+\pi)n} d(\Omega + \pi) \end{aligned}$$

- Obviously, $X_d(e^{j\Omega}) = X_e(e^{j(\Omega+\pi)}) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} X_c\left(j\left(\frac{\Omega+\pi-2k\pi}{T}\right)\right)$.
- If you were to draw a figure of $X_d(e^{j\Omega})$, it would be a graph of $X_c(e^{j\Omega})$ divided by T on the spectrum, then widened by T times ($-WT \leq \Omega \leq WT$), expanded by period 2π , and then shift π to the left or right.



2.2 Problem 2

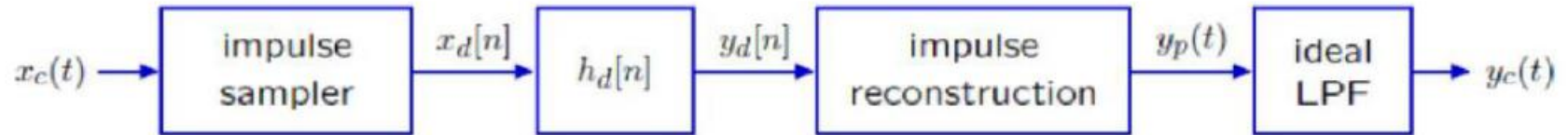
- (b) According to Nyquist's sampling theorem, $\omega_s \geq 2\omega_m$. As $\omega_s = 2\pi$, then $2\pi \geq \frac{2W}{T}$, $W \leq \frac{\pi}{T}$. The maximum value of W is $\frac{\pi}{T}$.
- (c) You can draw the diagram of the system like this. Use x to replace y .



2.3 Problem 3

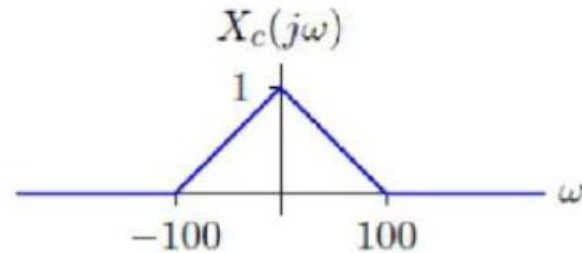
DT processing of CT signals

Sampling and reconstruction allow us to process CT signals using digital electronics as shown in the following figure.



The “impulse sampler” and “impulse reconstruction” use sampling interval $T = \pi/100$. The unit-sample function $h_d[n]$ represents the unit-sample response of an ideal DT low-pass filter with gain of 1 for frequencies in the range $-\frac{\pi}{2} < \Omega < \frac{\pi}{2}$. The “ideal LPF” passes frequencies in the range $-100 < \omega < 100$. It also has a gain of T throughout its pass band.

Assume that the Fourier transform of the input $x_c(t)$ is $X(j\omega)$ shown below.



Determine $Y_c(j\omega)$.



2.3 Problem 3

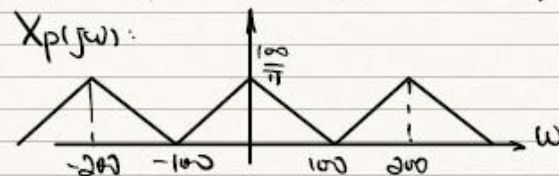
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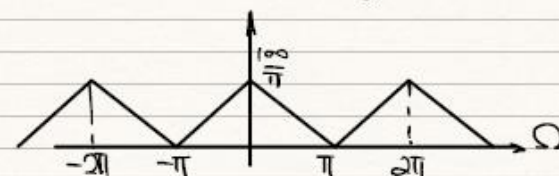
3. Sampling rate $\omega_s = \frac{2\pi}{T} = 200$

The sampled C-T signal is $x_p(t) = \sum_{n=-\infty}^{+\infty} x_a(nT) \cdot \delta(t-nT)$

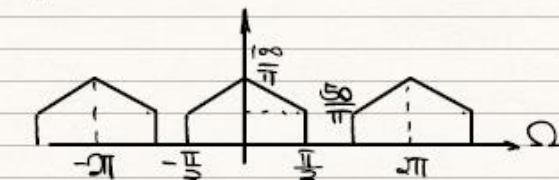
$X_p(j\omega)$:



$X_d(e^{j\Omega})$: A scaling is ω -domain from $X_p(j\omega)$ $\Omega = \omega T = \frac{\omega}{100}$

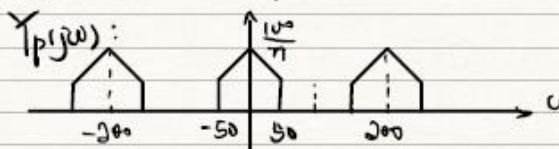


$Y_d(e^{j\Omega}) = X_d(e^{j\Omega}) \cdot H_d(e^{j\Omega})$, that is,

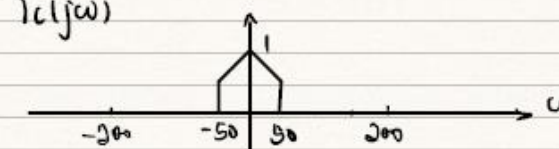


$y_p(t)$ is the impulse reconstructed form of $y_d(nT)$. $\omega = \frac{\Omega}{T} = \frac{100\Omega}{\pi}$

$Y_p(j\omega)$:



$Y_c(j\omega)$



2.4 Problem 4

- OWN Problem 7.6

7.6. In the system shown in Figure P7.6, two functions of time, $x_1(t)$ and $x_2(t)$, are multiplied together, and the product $w(t)$ is sampled by a periodic impulse train. $x_1(t)$ is band limited to ω_1 , and $x_2(t)$ is band limited to ω_2 ; that is,

$$X_1(j\omega) = 0, |\omega| \geq \omega_1,$$

$$X_2(j\omega) = 0, |\omega| \geq \omega_2.$$

Determine the *maximum* sampling interval T such that $w(t)$ is recoverable from $w_p(t)$ through the use of an ideal lowpass filter.

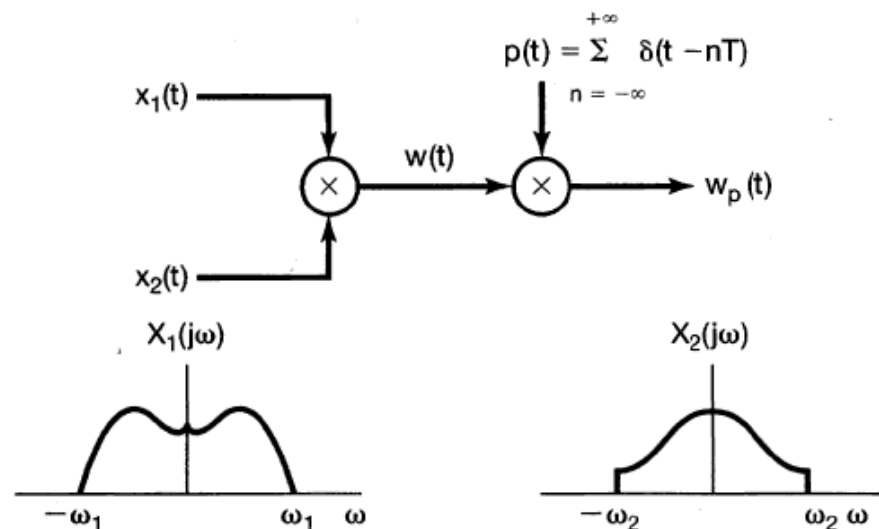


Figure P7.6



2.4 Problem 4

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- Consider the signal $w(t) = x_1(t)x_2(t)$. The Fourier transform $W(j\omega)$ of $w(t)$ is given by

$$W(j\omega) = \frac{1}{2\pi} [X_1(j\omega) * X_2(j\omega)]$$

- Since $X_1(j\omega) = 0$ for $|\omega| \geq \omega_1$ and $X_2(j\omega) = 0$ for $|\omega| \geq \omega_2$, we may conclude that $W(j\omega) = 0$ for $|\omega| \geq \omega_1 + \omega_2$.
- Consequently, the Nyquist rate for $w(t)$ is $\omega_s = 2(\omega_1 + \omega_2)$. Therefore, the maximum sampling method which would still allow $w(t)$ to be recovered is $T = 2\pi/\omega_s = \pi/(\omega_1 + \omega_2)$.



2.5 Problem 5

- OWN Problem 7.23

7.23. Shown in Figure P7.23 is a system in which the sampling signal is an impulse train with alternating sign. The Fourier transform of the input signal is as indicated in the figure.

- For $\Delta < \pi/(2\omega_M)$, sketch the Fourier transform of $x_p(t)$ and $y(t)$.
- For $\Delta < \pi/(2\omega_M)$, determine a system that will recover $x(t)$ from $x_p(t)$.
- For $\Delta < \pi/(2\omega_M)$, determine a system that will recover $x(t)$ from $y(t)$.
- What is the *maximum* value of Δ in relation to ω_M for which $x(t)$ can be recovered from either $x_p(t)$ or $y(t)$?

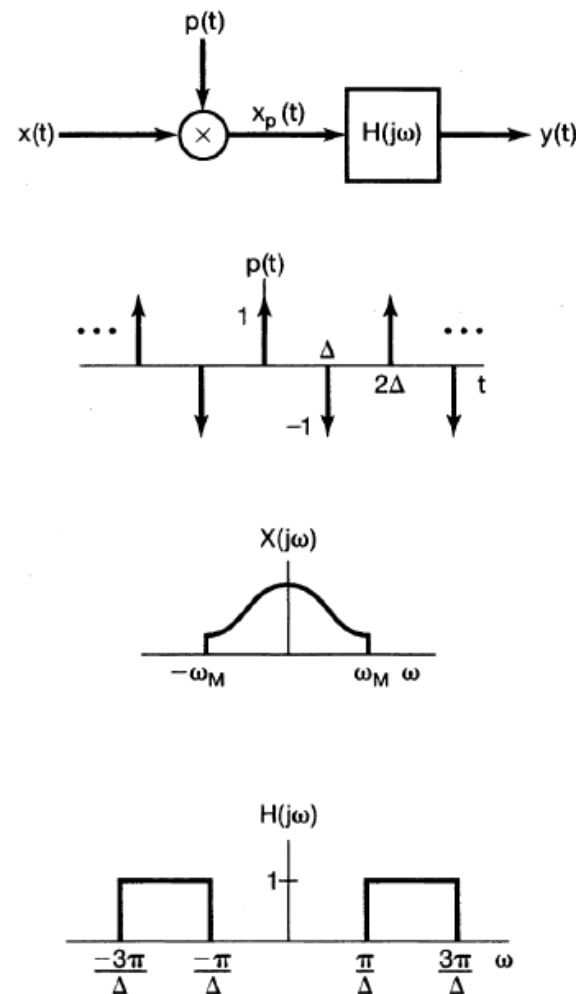


Figure P7.23



2.5 Problem 5

- (a) We may express $p(t)$ as

$$p(t) = p_1(t) - p_1(t - \Delta)$$

- Where $p_1(t) = \sum_{k=-\infty}^{\infty} \delta(t - k2\Delta)$. Now

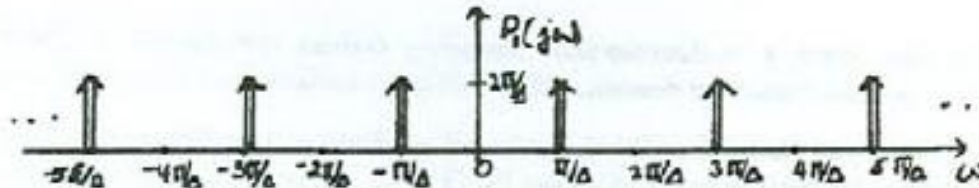
$$P_1(j\omega) = \frac{\pi}{\Delta} \sum_{l=-\infty}^{\infty} \delta\left(\omega - l\frac{\pi}{\Delta}\right)$$

- Therefore

$$P(j\omega) = P_1(j\omega) - e^{-j\omega\Delta}P_1(j\omega) = (1 - e^{-j\omega\Delta})P_1(j\omega) = \begin{cases} 2P_1(j\omega), & \omega = (2r + 1)\pi/\Delta, r \text{ is integer.} \\ 0, & \omega = 2r\pi/\Delta \end{cases}$$

- That is

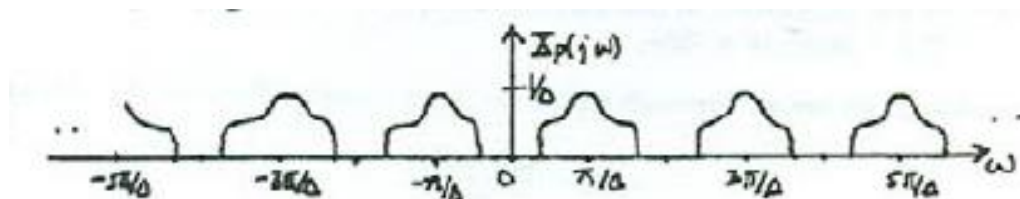
$$P(j\omega) = \frac{2\pi}{\Delta} \sum_{l=-\infty}^{\infty} \delta\left(\omega - (2l + 1)\frac{\pi}{\Delta}\right)$$



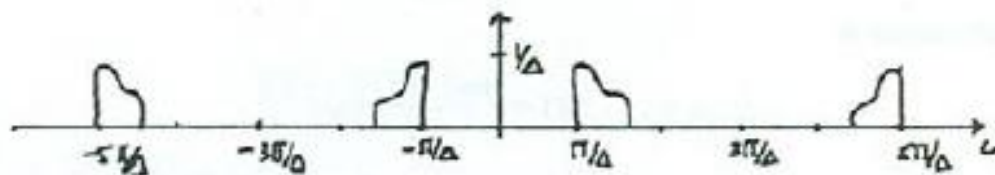
2.5 Problem 5

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- $x_p(t) = x(t)p(t)$, then $X_p(j\omega) = \frac{1}{2\pi} (X(j\omega) * P(j\omega)) = \frac{1}{\Delta} \sum_{l=-\infty}^{\infty} X \left(j \left(\omega - (2l + 1) \frac{\pi}{\Delta} \right) \right)$.



- $H(j\omega)$ is a bandpass filter, the Fourier transform of $y(t)$ is $Y(j\omega) = X_p(j\omega)H(j\omega)$.

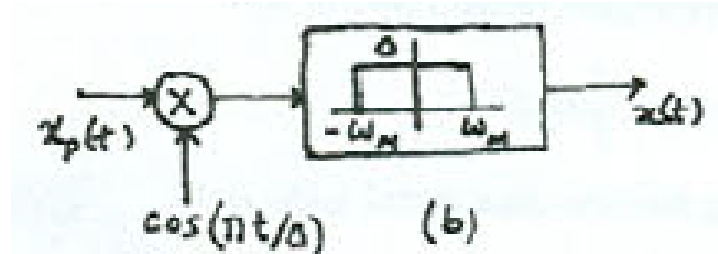


- Note that $y(t)$ isn't periodic, so you can't draw ellipses.

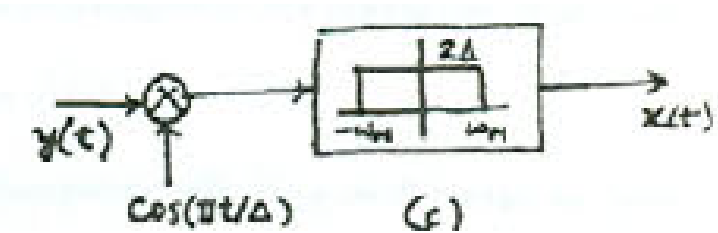


2.5 Problem 5

- (b) The system which can be used to recover $x(t)$ from $x_p(t)$ is like this. (You can only use one period for a periodic signal $x_p(t)$.)



- (c) The system which can be used to recover $x(t)$ from $y(t)$ is like this. Notice that the spectrum is cut in half, the amplitude of the lowpass filter should be twice that of part (b).



- (d) We see from the figures sketched in part (a) that aliasing is avoided when $\omega_M \leq \pi/\Delta$. Therefore, $\Delta_{\max} = \pi/\omega_M$.



2.6 A brief statement

- The deadline for Problem Set 8 has been adjusted to **June 23rd**, as the exam date is **June 26th**. There will be no lecture on problem sets after that, so Problem Set 8 **will not be reviewed in class**. The reference answers will be released on **June 24th** for review.





Done. Beg your criticism.

Presenter: Zhengyi Li

Date: June 11th